Type MS · Prepare cheek-cell or blood slides for a light microscope

Answer 1 of 6 · Question 7 part b · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

7 (a)

Part	Name of part	Function of part
X	cell membrane;	controls what enters/leaves cell;
Y	nucleus;	controls cell/stores genetic information;
Z	cytoplasm;	site of chemical reactions;

If they get any wrong in the part column, they can score marks for the correct function.

6

(b)

- scrape inside of cheek/cotton bud to remove cheek cells/eq.;
- place (scraping) on slide;
- smear/spread thinly;
- add stain/ methylene blue;
- place coverslip on top;
- remove excess stain;
- sterile/disinfect cotton bud;

4

(

(c)

- ribosomes;
- mitochondria;

accept any suitable structures such as ER,golgi,RER

ignore chromosomes

2

Total = 12

PAPER TOTAL - 90 MARKS

Type MS · Prepare cheek-cell or blood slides for a light microscope

Answer 1 of 6 · Question 7 part b · 4 marks

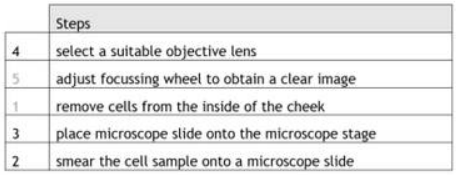
2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

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Type MS · Prepare cheek-cell or blood slides for a light microscope

Answer 2 of 6 · Question 3 part a.i · 2 marks

2020 NOV · Paper 01 (regional R) · find part a.i in this mark scheme extract

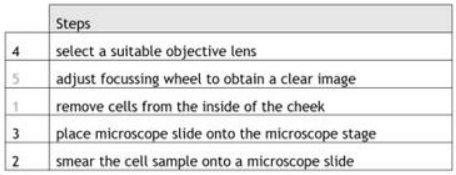
Number				
3	(a) (i)		One mark for 4 before 5. One mark for 3 before 2	2
	(ii)	more detail/organelles can be seen/organelles/named organelle seen more clearly (1)		1
	(iii)	reference no stain applied to specimen(1)		1
	(iv)	• reference to sterilising (named) equipment (1) • wash hands to avoid cross contamination(1) • incubate/store in sealed petri dish / container	allow valid sterilisation technique	2
	(b) (i)	10 × 40 = 400	Full marks for correct answer without working	1
	(ii)	reference to 1000 (1) 0.018 × 1000 = 18 (µm)	Full marks for correct final answer	2

Total Question 3 = 9 marks

Type MS · Prepare cheek-cell or blood slides for a light microscope

Answer 3 of 6 · Question 3 part a.iii · 1 mark

2020 NOV · Paper 01 (regional R) · find part a.iii in this mark scheme extract

3	(a) (i)		One mark for 4 before 5. One mark for 3 before 2	2
	(ii)	more detail/organelles can be seen/organelles/named organelle seen more clearly (1)		1
	(iii)	reference no stain applied to specimen(1)		1
	(iv)	• reference to sterilising (named) equipment (1) • wash hands to avoid cross contamination(1) • incubate/store in sealed petri dish / container	allow valid sterilisation technique	2
	(b) (i)	10 × 40 = 400	Full marks for correct answer without working	1
	(ii)	reference to 1000 (1) 0.018 × 1000 = 18 (µm)	Full marks for correct final answer	2

Total Question 3 = 9 marks

Type MS · Prepare cheek-cell or blood slides for a light microscope

Answer 4 of 6 · Question 6 part b · 3 marks

2023 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

6 (a)	• general shape with nucleus at centre; • nucleus labelled; • cytoplasm labelled; • cell membrane labelled;		4
(b)	• scrape/swab cells from (inside of) cheek • smear (thinly) over slide;/place sample on slide; • add stain/cover with coverslip;		3
(c)	• cheek cell shape irregular, trachea cell oblong; • cilia present in trachea, absent in cheek cell; • position of nucleus different;		3

Total 10 marks

Question number	Answer	Notes	Marks

Type MS · Prepare cheek-cell or blood slides for a light microscope

Answer 5 of 6 · Question 6 part b.i · 3 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
6	(a)	(i)	phagocytosis;		1
		(ii)	• enclosed bacterium fuses with lysosome; • enzymes break down /digests bacterium; • remains of bacterium removed from white blood cell;		3
		(iii)	• label/identify/complementary to bacterium/antigens presented; • so response quicker if new infection/better immune response;		2
	(b)	(i)	any three from • prick skin to obtain blood; • use a sterile needle/pin • smear/spread blood on slide; • ensure smear is thin; • add stain; • allow to dry;		3
		(ii)	any four from • red blood cells would be seen as circle/not in 3D; • white blood cells would be larger than red/rbc too large; • wouldn't see platelets; • because too small for light microscope;		4
		(iii)	• no risk of infection; • e.g. AIDS;	R contaminate	2

Total 15

Type MS · Prepare cheek-cell or blood slides for a light microscope

Answer 5 of 6 · Question 6 part b.i · 3 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Prepare cheek-cell or blood slides for a light microscope

Answer 6 of 6 · Question 7 part b · 5 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

7 (a) (i)		• 2nd set of parents have genotypes I^BI^O and I^BI^O/both parents are I^BI^O; • give rise to child I^OI^O which is M; • 1st set of parents have genotypes I^OI^O and I^AI^B; • can give rise to a child I^AI^O which is N;	Allow 1 mark if the child is matched to the correct parent.	4
	(ii)	• parental genotypes $I^BI^O \times I^BI^O$; • offsprings are I^BI^B 2 I^BI^O I^OI^O; • 75%/0.75/3:1/3 in 4;	accept parental gametes for mp1 if BO and BO parental genotypes are used they can get ecf for MP2. accept offsprings in genetic cross	3
(b)		• (prick finger) with pin/needle; • sterile; • (smear) blood on slide; • thinly; • dry; • stain;	accept example of sterile Accept MP5 and MP6 in either order	5
				Total =12

Type MS · Prepare cheek-cell or blood slides for a light microscope

Answer 6 of 6 · Question 7 part b · 5 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

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